

A Second Opinion for the Portfolio

How a New Forecasting Model Changes Cross-Asset Allocation Decisions

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Abstract

New forecasting models are often judged by prediction error. A portfolio, however, holds positions rather than forecasts. This paper studies that distinction in Forecast-to-Rank Allocation (F2R), a monthly cross-asset process that ranks 11 U.S.-listed ETFs, selects the Top-4, and assigns a 25% target weight to each. We compare the incumbent four-model ensemble, Chronos-2, and matched hybrids across 112 monthly signal dates under the same portfolio, execution, and cost rules.

Chronos-2 has a higher point-forecast RMSE (0.0570) than the conventional models (about 0.0550–0.0553), yet its standalone Top-4 portfolio reaches a 15.5% CAGR and 0.962 Sharpe ratio versus 14.9% and 0.891 for the incumbent. A three-model hybrid with a 20% Chronos-2 contribution reaches 17.1% CAGR and 0.987 Sharpe with lower cumulative turnover. The same 20% contribution is associated with only a 0.61-percentage-point CAGR increase when added to the incumbent ensemble but a 2.12-point increase after Random Forest is removed. Holding the models and 80/20 weights fixed, rank-based integration also exceeds standardized-forecast integration by 2.53 percentage points of CAGR with nearly identical turnover.

These are descriptive historical contrasts, not causal estimates. The exercise is a retrospective reconstruction using today’s forecasting technology; Chronos-2 did not exist throughout 2017–2026, its pretraining corpus is not audited for temporal overlap, and the main hybrid was developed within the same historical research program rather than on an untouched holdout. The practical lesson is narrow: judge a new forecasting model by the portfolio decisions it changes, how its forecasts are integrated, and the consequences of those decisions—not only by average forecast error.

1. The Investment Question

An investor deciding whether to use a new forecasting model faces a different problem from a forecaster choosing the model with the smallest prediction error. Forecast accuracy matters, but it is only the first link in a longer chain. The forecast must change a ranking, the ranking must change a position, and the changed position must improve the realized portfolio after turnover, costs, and risk are taken into account.

This distinction is familiar in decision theory and predict-then-optimize research: a statistically better estimate does not automatically produce a better downstream decision (Granger and Pesaran, 2000; Bertsimas and Kallus, 2020; Elmachoub and Grigas, 2022). Asset-allocation research has also shown that machine-learning forecasts and forecast combinations can have economic value (Kynigakis and Panopoulou, 2022). The contribution here is therefore narrower: a matched case showing how the composition of a forecasting ensemble and the interface used to combine its forecasts can alter actual portfolio decisions. The practical question is:

Does the new model improve the portfolio decision, or only the forecast?

***Takeaway.** A new forecasting model does not need to win the forecasting contest to earn a role in the portfolio process. It needs to change the decisions that matter.*

2. The F2R Decision Process

Forecast-to-Rank Allocation (F2R) is a monthly cross-asset process that evaluates 11 U.S.-listed ETFs representing liquid asset exposures, converts model forecasts into relative rankings across the 11 ETFs, and allocates 25% target weight to each of the four highest-ranked ETFs. The present study keeps that portfolio mapping, universe, execution timing, and transaction-cost treatment fixed and changes the forecasting architecture upstream. The universe spans U.S. large- and small-cap equities, international equities, real estate, U.S. Treasuries, broad commodities, and gold; Appendix A lists the ETF proxies.¹

The rank interface is important because the models do not naturally speak the same numerical language. Rather than averaging raw forecasts that may differ in scale and calibration, F2R can first ask a common question: how does each model rank the 11 assets? Those ranks are then combined before the Top-4 portfolio is selected. This makes heterogeneous forecasts comparable, but it also discards information about forecast magnitude. The paper therefore treats rank integration as an empirical portfolio-design choice rather than an assumption.

The Top-4 cutoff makes small ranking changes economically meaningful. If three holdings remain the same but the fourth- and fifth-ranked assets exchange places, one quarter of the target portfolio changes. This selection boundary is where a new forecasting model can provide a useful “second opinion.”

3. What Exactly Changed in the Forecasting Ensemble?

The incumbent F2R ensemble combines four conventional machine-learning models. Two are regularized linear models (Ridge and Elastic Net), and two are tree-based nonlinear models (Random Forest and XGBoost). Their asset rankings are combined before the Top-4 portfolio is selected.

Chronos-2 differs in its training paradigm. It is a pretrained time-series forecasting model: rather than being fitted from scratch only within the F2R sample, it brings forecasting structure learned during a much broader pretraining process. In this experiment it is applied without additional task-specific training on the F2R sample (Ansari, Stella, et al., 2024; Ansari, Shchur, et al., 2025). At each historical signal date, the portfolio process is restricted to price-only inputs available at that date. The models are nevertheless not informationally identical in a broader sense. The conventional models use engineered price features and are fitted within F2R; Chronos-2 consumes a sequence of daily log returns and brings prior information from pretraining. This study does not audit the contents or temporal overlap of the Chronos-2 pretraining corpus. Macroeconomic covariates are deliberately excluded so that the model-adoption question is not mixed with a change in the portfolio-side information set.

The analysis uses six recurring configurations. The percentages below are weights in the *model combination*, not allocations of portfolio capital.

Configuration	Forecasting components	Integration
Incumbent ensemble	Ridge, Elastic Net, Random Forest, XGBoost	Rank
Three-model ensemble	Ridge, Elastic Net, XGBoost	Rank
Chronos-2 alone	Chronos-2	Rank
Chronos + incumbent	Incumbent ensemble + 20% Chronos-2	Rank
Selected hybrid	Three-model ensemble + 20% Chronos-2	Rank
Standardized hybrid	Same as selected hybrid, same 80/20 weights	Standardized forecast magnitude

The label *selected hybrid* identifies the configuration carried forward for the paper’s main comparison; it does not imply that the 20% contribution was chosen by a prespecified optimizer or is an estimated optimum.

¹See Sungkyu Lee (2026), *The Decision Value of Price and Macro Information: Benchmark-Relative Evidence in Multi-Asset Allocation Using Historical-Vintage FRED-MD Data*, SlackQuant Investment Research Series, SSRN Abstract ID 7340100. That earlier study asks whether price or macro information adds decision value. The present paper keeps the portfolio-side information set price-only and instead studies model adoption and forecast integration.

The controlled comparison spans 112 monthly signal dates from March 2017 through June 2026. Every variant uses the same 11-ETF universe, Top-4 portfolio rule, execution-accounting convention, and transaction-cost framework. Two reconstruction checks anchor the comparison: the reconstructed incumbent decisions match the stored incumbent decision history for all 112 signal dates, and the reconstructed selected-hybrid decisions match the corresponding stored hybrid decision history for all 112 dates. These checks verify that the historical portfolio mechanics are held fixed while the forecasting configuration changes.

Experimental design in brief. The 112 signal dates run from March 2017 through June 2026; portfolio metrics use 2,327 daily observations from May 1, 2017 through August 3, 2026. The first March 2017 signal executes in early April, but reported performance starts May 1 to exclude that initial partial month; the series extends through August 3 to complete the holding interval associated with the final June 2026 signal. Signals use the last common U.S. market close at or before month-end and execute at the first subsequent common close. Holdings drift until rebalance; one-way turnover is half the total absolute change between drifted pre-trade and target weights (the half- L_1 convention), cost is 10 basis points on each side, and Sharpe uses daily net returns with a zero risk-free rate and $\sqrt{252}$ annualization. Forecast error for every model is measured against the same next holding-period return target aligned to the execution clock. The model inputs are not identical: Chronos-2 uses the preceding 252 trading days of log returns and converts the median forecast path after execution into its point signal, while the conventional models use engineered price features. The rank-versus-standardized comparison fixes forecasts, models, 80/20 weights, Top-4 selection, execution, and costs and changes only the integration rule. Exact tie rules, frozen panels, and regeneration code are in the versioned research supplement.

4. Forecast Accuracy Gives an Incomplete First Impression

The first comparison puts point-forecast accuracy beside the investment outcome produced when each forecasting model supplies the asset ranking on its own. Figure 1 shows why the two scorecards should not be treated as interchangeable.

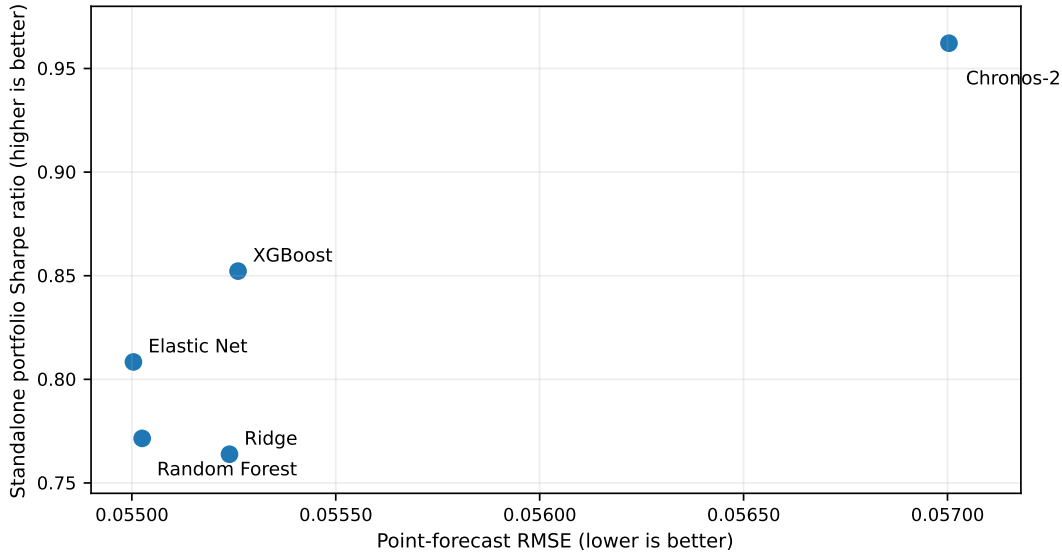


Figure 1. Forecast accuracy and standalone portfolio performance do not rank the models the same way. The horizontal axis reports root mean squared error (RMSE) across the common forecast panel, where lower is better. The vertical axis reports the Sharpe ratio, with the risk-free rate set to zero, of the Top-4 portfolio formed from each model's rankings under the same execution, cost, and accounting rules. The comparison uses the same 112 monthly signal dates. The figure is descriptive; it does not imply that higher RMSE causes better portfolio performance.

Chronos-2 has the highest reported RMSE among the five standalone forecasting models at 0.0570. Elastic Net, Random Forest, Ridge, and XGBoost all cluster near 0.0550–0.0553. If the adoption decision stopped at point-forecast error, Chronos-2 would not be the obvious choice.

The portfolio ordering is different. Chronos-2 alone produces a 15.49% CAGR and a 0.962 Sharpe ratio; the incumbent four-model ensemble produces 14.92% and 0.891 under the same accounting framework.

This does not make forecast accuracy irrelevant. It means that the loss function used to rank forecasts is not the portfolio objective itself. Decision-focused prediction emphasizes the same broad distinction: the forecasting loss and the downstream decision objective need not coincide (J. Lee et al., 2025). F2R does not train the models on a portfolio objective here; it shows the divergence in a controlled predict-then-rank setting after forecasts are translated into positions.

5. Where Did the Portfolio Improvement Come From?

Relative to the incumbent four-model ensemble, the selected hybrid has a 2.21-percentage-point higher CAGR and a 0.096 higher Sharpe ratio, while cumulative one-way turnover falls from $27.55\times$ to $20.51\times$ portfolio value over the reported history. One-way turnover is the half- L_1 distance between drifted pre-trade and target weights; a value of 1.0 corresponds to one-way replacement equal to 100% of portfolio value.

Because the selected hybrid both removes Random Forest and adds Chronos-2, the headline comparison alone cannot attribute the improvement. Figure 2 uses matched historical contrasts to separate those changes.

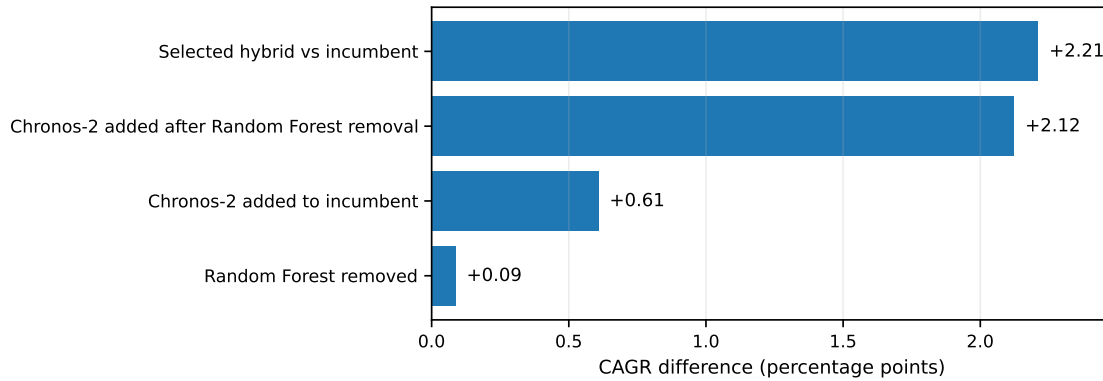


Figure 2. The improvement is not explained by a single ensemble change. Bars report CAGR differences for matched historical comparisons under the same portfolio-side data, portfolio, execution, and cost rules. Random Forest removal alone corresponds to a 0.09-percentage-point higher CAGR; the same 20% Chronos-2 model-combination contribution corresponds to a 0.61-point increase in the incumbent ensemble and a 2.12-point increase after Random Forest removal. The 2.21-point overall comparison reflects both changes. These are descriptive contrasts, not causal estimates.

Removing Random Forest alone is associated with a 0.09-percentage-point higher CAGR and a 0.022 lower Sharpe ratio. Adding a 20% Chronos-2 model-combination contribution to the unchanged four-model ensemble is associated with a 0.61-point higher CAGR and a 0.023 higher Sharpe ratio. After Random Forest removal, the same 20% contribution is associated with a 2.12-point higher CAGR and a 0.118 higher Sharpe ratio.

That pattern does not identify a universal “Chronos effect.” In this experiment, the incremental value of the new forecasting view can depend on the ensemble it joins: the same model contribution is modest in the incumbent ensemble and materially larger after the conventional ensemble is reconfigured.

6. Where the Models Meet Matters

The selected hybrid combines model opinions after converting forecasts into relative asset rankings. The matched alternative uses the same three conventional models, the same Chronos-2 forecasts, and the same 80/20 model-combination weights, but instead standardizes each model’s forecasts across the 11 ETFs before combining them. Standardization puts forecasts on a common scale while retaining information about relative magnitude.

The rank-based variant has a 2.53-percentage-point higher CAGR and a 0.163 higher Sharpe ratio with nearly identical cumulative turnover. This is not universal proof that rank combination dominates. It shows that the *interface* between forecasting and portfolio construction can matter when models produce outputs on different scales.

Table 1. Same forecasts and model-combination weights, different integration interface.

	CAGR	Sharpe	Max DD	Turnover ($\times$ NAV)
Rank-based hybrid	17.13%	0.987	-21.97%	20.51
Standardized-forecast hybrid	14.60%	0.824	-24.55%	20.48

Rank translation can reduce scale and calibration differences across heterogeneous models, but it discards information about forecast magnitude. The current F2R comparison favors the rank-based interface; whether that trade-off generalizes remains an open empirical question.

7. The Second Opinion Matters When It Changes the Top-4

The selected hybrid does not create a different portfolio every month. Across 112 signal dates, its Top-4 set differs from the incumbent in 58 months (51.8%) and is identical in the remaining 54. Its average Top-4 Jaccard similarity with the incumbent is 0.783, where 1 means identical selections; similarity is higher against the three-model conventional ensemble, at 0.875. In other words, the new architecture usually leaves most of the portfolio view intact and matters when it changes which assets cross the Top-4 cutoff.

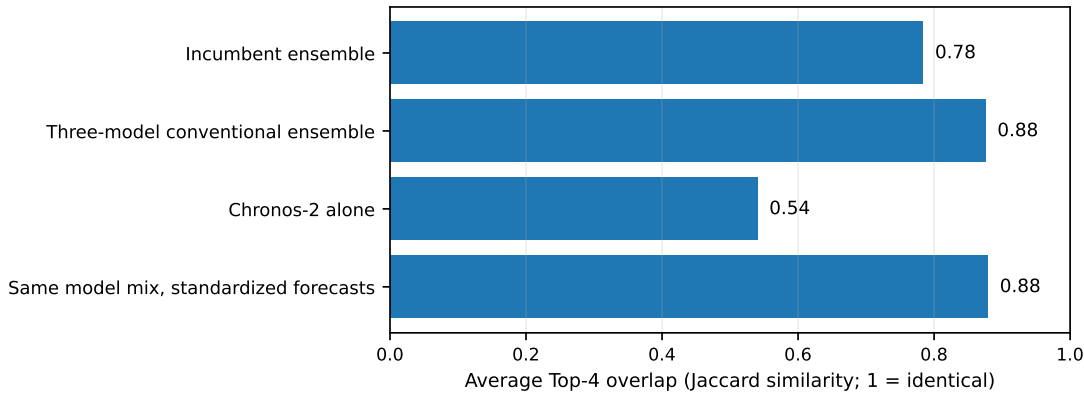


Figure 3. Most monthly Top-4 choices remain similar. Bars show average Jaccard overlap between the selected hybrid and each alternative across 112 monthly signal dates. A value of 1 means identical Top-4 sets. The figure measures decision similarity, not performance contribution.

Similarity alone does not tell us whether the changed months matter for wealth. We therefore attribute the full hybrid-versus-incumbent terminal wealth difference back to each monthly signal decision using an exact decomposition of the cumulative performance gap in log wealth. Gross-return differences are assigned to the target active over the corresponding holding interval, and transaction-cost differences to the incoming rebalance. The components reconcile exactly to the two reconstructed portfolio histories.

Those 58 changed-selection months account for 99.4% of the net relative log-wealth gap between the hybrid and the incumbent. Because both variants target the same equal-weight Top-4 portfolio when their selections coincide, the gross-return contribution in unchanged-selection months is zero by construction; the small residual comes only from path-dependent trading-cost differences. The 99.4% figure is therefore an accounting and transmission diagnostic, not independent evidence that the new model is useful.

Of the 58 changed decisions, 36 contribute positively and 22 negatively. The largest favorable decision accounts for 10.7% of total positive changed-decision contribution. The five largest favorable decisions account for 35.2% of positive contribution but 71.3% of the net relative log-wealth gap, because negative changed decisions offset a substantial share of the gains. The result is not attributable to a single substitution, but the net advantage remains concentrated.

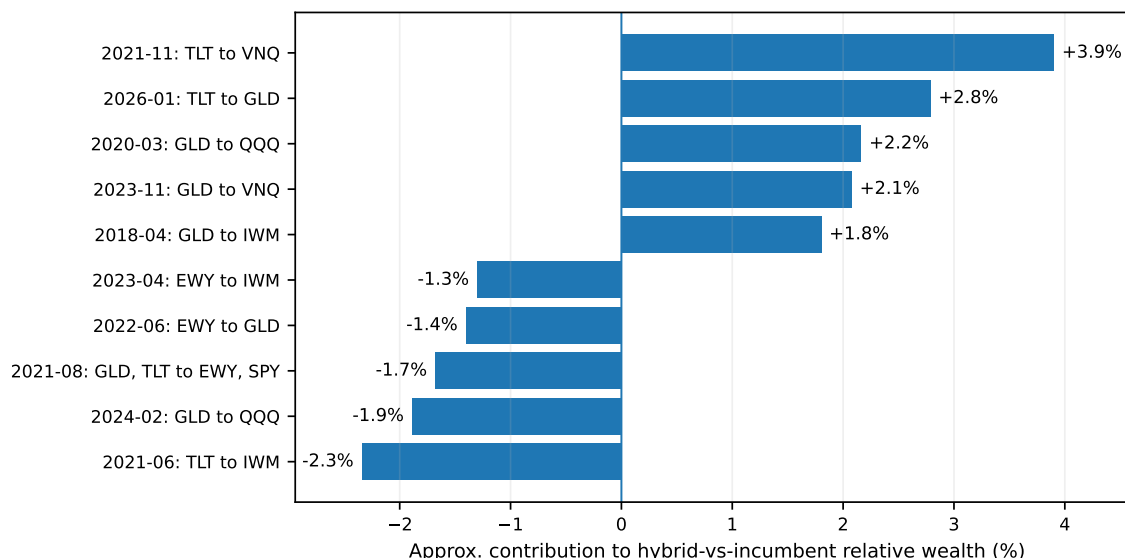


Figure 4. The changed decisions include both winners and losers. The figure shows the five largest favorable and five largest unfavorable changed Top-4 decisions. Each label gives the signal month and the incumbent asset or assets replaced by the hybrid. Bar labels convert each event's log-wealth contribution to an approximate relative-return percentage for readability; exact reconciliation is performed in log wealth. This is a post-hoc mechanism diagnostic, not independent out-of-sample evidence.

The mechanism evidence therefore supports a narrow version of the “second opinion” interpretation. The historical relative advantage is transmitted almost entirely through months when the new architecture changes the Top-4 set. It is not a one-event story, but the net contribution is concentrated in a relatively small set of favorable decisions. The attribution does not establish that Chronos-2 alone caused the advantage, nor does it remove retrospective model-selection risk.

8. A Stronger Backtest Is Not Automatically the Better Choice

The selected hybrid is not the highest-return candidate in the historical comparison. A separate candidate retains the incumbent four-model ensemble and gives Chronos-2 50% of the model-combination weight. It reaches a 17.63% CAGR and a 1.042 Sharpe ratio, versus 17.13% and 0.987 for the selected hybrid. Its maximum drawdown is also deeper at -24.27%, versus -21.97%. This four-model 50% candidate is different from the three-model weight-sensitivity family shown below.

Figure 5 reports that separate three-model comparison. Its 50% configuration has a 16.72% CAGR, not the 17.63% of the four-model candidate above; the existing 25% configuration has a 16.20% CAGR. Across the displayed three-model configurations, CAGR does not increase monotonically with the Chronos-2 contribution. These historical comparisons do not identify an optimal weight or validate a general model-adoption rule.

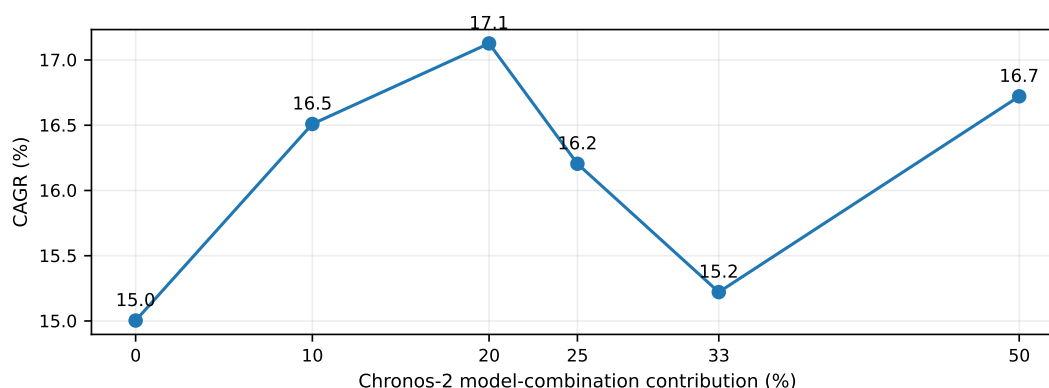


Figure 5. The historical result is not a smooth optimization curve. Within the same three-model conventional ensemble, the Chronos-2 model-combination contribution is varied across 0%, 10%, 20%, 25%, 33%, and 50%. CAGR is non-monotonic, so the selected 20% contribution should not be interpreted as an estimated optimum.

The selected 20% configuration therefore represents one observed return, drawdown, and turnover trade-off rather than an estimated optimum or a validated adoption policy. A higher historical return in one configuration is useful evidence, but it is not by itself a rule for choosing how much weight to give a new forecasting model.

Takeaway. *Model adoption is a portfolio-design decision. Evaluate forecast quality, changed decisions, turnover, risk outcomes, and failure modes together; do not choose a model simply because one historical backtest cell is the maximum.*

9. What the Evidence Does Not Prove

The most important limitation is temporal. Chronos-2 did not exist throughout the 2017–2026 historical window, so this exercise does not claim that an investor could literally have run it in 2017. It is a retrospective counterfactual: today’s model is inserted into the historical F2R process using only the price data available to that process at each signal date and the same implementation rules. Because Chronos-2 is pretrained, however, this does not imply identical training information or prior knowledge across models; the study does not audit the contents or temporal overlap of its pretraining corpus. Accordingly, the Chronos-2 results should not be read as a real-time historical out-of-sample track record.

Second, the selected hybrid was developed inside the same broader historical research program. Ablation clarifies which design changes are associated with the result, but it does not create a pristine untouched holdout sample or remove model-selection risk. Third, the evidence comes from one 11-asset universe, one Top-4 decision rule, and one pretrained model family. Fourth, the rank-versus-standardized-forecast comparison is informative inside this system but is not a universal test of forecast representation.

The attribution was conducted after the main comparison was known and remains a post-hoc mechanism diagnostic. The five largest favorable decisions account for 71.3% of the net relative log-wealth gap, so concentration remains important even though no single substitution explains the result. Future evidence should be prospective and reported separately from this reconstruction; another universe or decision rule would test portability. A separate follow-up can examine macroeconomic covariates without mixing that information-set question into the present price-only model-adoption test.

10. Practical Model-Adoption Checklist

The F2R experiment suggests a compact audit that an investor, researcher, or investment team can apply beyond this particular system:

1. **Start with the portfolio decision.** What position, ranking, or allocation is the new model supposed to improve?
2. **Hold the investment setup fixed where possible.** Keep the portfolio-side data, portfolio rule, execution rule, and cost treatment fixed, and disclose differences in pretraining or input representation.
3. **Report forecast and portfolio metrics side by side.** Better prediction error and better investment outcomes need not rank models in the same order.
4. **Compare one change at a time.** Use matched comparisons to test addition, removal, reconfiguration, and the forecast-to-portfolio interface rather than relying on one headline backtest.
5. **Check what changed and where the gains came from.** In a strategy that holds only the highest-ranked assets, a small rank change near the cutoff can move a large portfolio weight, and a few decisions can dominate the net result.
6. **Do not turn one historical result into a universal rule.** A historical trade-off can motivate further use or testing without establishing an optimal configuration that will generalize.

11. Conclusion

The most informative result is not simply that Chronos-2 has a worse RMSE but a stronger standalone portfolio. The same 20% Chronos-2 contribution is associated with a much larger historical improvement after the conventional ensemble is reconfigured than when it is appended to the incumbent ensemble. A second matched comparison shows that, with the same forecasts and model-combination weights, rank-based and standardized-forecast integration can produce materially different portfolio outcomes with almost identical turnover.

The evidence therefore argues against treating model adoption as a forecasting leaderboard. In the retrospective attribution, the selected hybrid changes the Top-4 set in roughly half of the evaluated decisions, and those changed selections transmit essentially the entire net relative log-wealth gap by construction of the equal-weight Top-4 portfolio. The net advantage is not a one-event result, but it is concentrated: the five largest favorable decisions account for 71.3% of the net gap. The more useful question is whether a new model supplies a sufficiently different view to change which assets make the portfolio, and whether the resulting decisions remain attractive after implementation costs and risk are assessed.

For investors and investment researchers, the durable principle is simple: **do not ask only whether a new model forecasts better. Ask what portfolio decision it changes, why it changes it, and whether that changed decision earns its keep.**

A. Asset-Universe Reference

Table 2 lists the 11 ETF proxies used in the model-adoption comparison. The broader universe-selection rationale is documented in the earlier F2R study cited in Section 2 and is not repeated here.

Table 2. F2R asset universe used in the model-adoption comparison.

Ticker	Economic exposure
SPY	U.S. large-cap equities (S&P 500)
QQQ	U.S. Nasdaq-100 equities
IWM	U.S. small-cap equities (Russell 2000)
EFA	Developed-market equities outside the U.S. and Canada
EEM	Emerging-market equities
EWY	South Korean equities
VNQ	U.S. listed real estate / REITs
IEF	Intermediate U.S. Treasuries (7–10 year)
TLT	Long-duration U.S. Treasuries (20+ year)
DBC	Broad commodities
GLD	Gold

References

- Ansari, Abdul Fatir, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, Huibin Shen, Lorenzo Stella, Xiyuan Zhang, Mononito Goswami, Shubham Kapoor, Danielle C. Maddix, Pablo Guerron, Tony Hu, Junming Yin, Nick Erickson, Prateek Mutalik Desai, Hao Wang, Huzefa Rangwala, George Karypis, Yuyang Wang, and Michael Bohlke-Schneider (2025). “Chronos-2: From Univariate to Universal Forecasting”. In: *arXiv preprint arXiv:2510.15821*.
- Ansari, Abdul Fatir, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang (2024). “Chronos: Learning the Language of Time Series”. In: *arXiv preprint arXiv:2403.07815*.
- Bertsimas, Dimitris and Nathan Kallus (2020). “From Predictive to Prescriptive Analytics”. In: *Management Science* 66.3, pp. 1025–1044. DOI: [10.1287/mnsc.2018.3253](https://doi.org/10.1287/mnsc.2018.3253).
- Elmachtoub, Adam N. and Paul Grigas (2022). “Smart “Predict, then Optimize””. In: *Management Science* 68.1, pp. 9–26. DOI: [10.1287/mnsc.2020.3922](https://doi.org/10.1287/mnsc.2020.3922).
- Granger, Clive W. J. and M. Hashem Pesaran (2000). “Economic and Statistical Measures of Forecast Accuracy”. In: *Journal of Forecasting* 19.7, pp. 537–560. DOI: [10.1002/1099-131X\(200012\)19:7<537::AID-FOR769>3.0.CO;2-G](https://doi.org/10.1002/1099-131X(200012)19:7<537::AID-FOR769>3.0.CO;2-G).
- Kynigakis, Iason and Ekaterini Panopoulou (2022). “Does Model Complexity Add Value to Asset Allocation? Evidence from Machine Learning Forecasting Models”. In: *Journal of Applied Econometrics* 37.3, pp. 603–639. DOI: [10.1002/jae.2885](https://doi.org/10.1002/jae.2885).
- Lee, Junhyeong, Haeun Jeon, Hyunglip Bae, and Yongjae Lee (2025). “Return Prediction for Mean-Variance Portfolio Selection: How Decision-Focused Learning Shapes Forecasting Models”. In: *Proceedings of the 6th ACM International Conference on AI in Finance*, pp. 114–122. DOI: [10.1145/3768292.3770423](https://doi.org/10.1145/3768292.3770423).
- Lee, Sungkyu (2026). “The Decision Value of Price and Macro Information: Benchmark-Relative Evidence in Multi-Asset Allocation Using Historical-Vintage FRED-MD Data”. SlackQuant Investment Research Series. SSRN Abstract ID 7340100.